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VisionGuard: AI-Powered Driver Drowsiness Monitoring

English script

Good morning everyone. Today we are presenting **VisionGuard**, an AI-powered driver drowsiness monitoring system.
The purpose of this project is not to make a medical diagnosis or replace driver responsibility.
Instead, our goal is to detect visible signs such as eye closure and yawning, then convert them into transparent warning-candidate states.
In this presentation, we will explain the problem, our system architecture, datasets, models, fusion logic, and the working interface.

中文脚本

大家好，今天我们展示的项目是 **VisionGuard**，一个基于 AI 的驾驶员疲劳监测系统。
这个项目的目标不是做医学诊断，也不是替代驾驶员自己的判断，而是检测一些可见的疲劳相关行为，比如闭眼和打哈欠，然后把这些证据转换成清晰、可解释的预警候选状态。
接下来我们会介绍问题背景、系统架构、数据集、模型、融合逻辑，以及最终的系统界面。

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Drowsiness is a multi-cue, time-based signal

English script

The key design idea starts from one observation: drowsiness is not a single-frame problem.
A closed eye in one frame may simply be a blink. A single yawn may be temporary. A poor camera angle may also create misleading visual evidence.
So instead of judging one snapshot directly, we split the problem into specialist visual signals. The eye model focuses on eye closure, the mouth model focuses on yawning, and the fusion layer checks whether these signals persist or co-occur over time.
This makes the system more transparent and easier to review.

中文脚本

这个项目的核心设计思想来自一个现实问题：疲劳不是一个单帧图像就能判断的问题。
一帧闭眼可能只是眨眼，一次打哈欠也可能只是暂时的行为，摄像头角度不好也可能产生错误视觉证据。
所以我们没有直接用一张图判断“疲劳或不疲劳”，而是把问题拆成多个专业视觉信号。眼部模型负责检测闭眼，嘴部模型负责检测打哈欠，融合层再判断这些信号是否持续出现，或者是否在时间上共同出现。
这样系统会更透明，也更容易解释和复查。

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System Architecture

English script

This slide shows the full system architecture.
At the top, the user interacts with the system through the web interface, including Live Monitor, Video Upload Analysis, History, and Insights.
After input is captured, the preprocessing layer uses face and landmark detection to extract regions of interest, mainly the eyes and mouth.
Then two specialist CNN models generate two probability signals: `p_eye_closed` and `p_yawn`.
These signals are not used in isolation. They are passed into a temporal rule-based fusion layer, which produces warning-candidate states and stores compact summaries for review.

中文脚本

这一页展示的是系统整体架构。
最上层是用户界面，包括实时监测、视频上传分析、历史记录和分析洞察页面。
当系统获得输入之后，预处理层会通过人脸和关键点检测，提取关键区域，主要是眼睛区域和嘴部区域。
之后，两个专门的 CNN 模型会分别输出两个概率信号：`p_eye_closed` 表示闭眼概率，`p_yawn` 表示打哈欠概率。
这些信号不会被单独直接当成最终结论，而是进入基于时间规则的融合层，生成预警候选状态，并保存简洁的记录用于后续查看。

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From raw datasets to a live, explainable system

English script

Our workflow has two main parts: offline training and online runtime.
In the offline stage, we inspected the datasets, generated eye and mouth crops, created subject-level splits, trained specialist models, and selected the final checkpoints.
In the runtime stage, the system captures a webcam frame or uploaded video frame, extracts the eye and mouth regions, runs specialist inference, and then applies temporal fusion.
The final output is not just a model prediction. It includes warning intervals, keyframes, history records, and visual evidence for interpretation.

中文脚本

我们的工作流程主要分为两个部分：离线训练和在线运行。
在离线阶段，我们检查数据集，生成眼部和嘴部裁剪图，进行按 subject 划分的数据拆分，训练专门模型，并选择最终使用的模型 checkpoint。
在在线运行阶段，系统会从摄像头或上传视频中获取帧，提取眼睛和嘴部区域，运行两个专门模型，然后进行时间融合判断。
最终输出不是一个简单的模型预测，而是包含预警时间段、关键帧、历史记录和可视化证据，方便解释和复查。

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Dataset Strategy

English script

We used two complementary datasets.
For the eye module, we used the MRL Eye dataset, which contains about **84,898 eye images** across **37 subjects**.
For the mouth and yawn module, we used YawDD and YawDD+ data, producing about **64,202 mouth crops** across **29 subjects**.
A very important point is that we used subject-level splitting. This reduces identity leakage, because images from the same person should not appear in both training and testing sets.
This makes the evaluation more realistic than a random image-level split.

中文脚本

我们使用了两个互补的数据集。
眼部模块使用的是 MRL Eye 数据集，其中包含大约 **84,898 张眼部图像**，来自 **37 个 subject**。
嘴部和打哈欠模块使用的是 YawDD 和 YawDD+ 数据，最终生成大约 **64,202 张嘴部裁剪图**，来自 **29 个 subject**。
这里非常重要的一点是，我们使用的是 subject-level split，也就是按人物划分训练集和测试集。这样可以减少身份泄漏，因为同一个人的图像不应该同时出现在训练集和测试集中。
这比随机按图片划分更接近真实评估。

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YawDD mouth preprocessing pipeline

English script

This slide focuses on the YawDD mouth preprocessing pipeline.
First, we reconstructed frames from the original YawDD driving videos using YawDD+ annotation frame indices.

Then, instead of directly trusting the original bounding boxes, we used MediaPipe Face Mesh lip landmarks to generate mouth crops.
When landmark detection was not reliable, a lower-face fallback crop was used.
Finally, the processed samples were split by subject.
This step is important because the mouth model depends heavily on clean and consistent mouth-region input.

中文脚本

这一页重点讲 YawDD 嘴部数据的预处理流程。
首先，我们根据 YawDD+ 标注中的 frame index，从原始 YawDD 驾驶视频中重建对应的视频帧。
然后，我们没有直接依赖原始 bounding box，而是使用 MediaPipe Face Mesh 的嘴唇关键点来生成嘴部裁剪图。
如果关键点检测不稳定，就使用下半脸区域作为 fallback crop。
最后，所有处理后的样本按照 subject 进行划分。
这一步很关键，因为嘴部模型的效果很大程度上取决于输入的嘴部区域是否干净、稳定、一致。

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Specialist Models

English script

Each component in VisionGuard owns one visible behavior.
For the eye module, the selected model is **MobileNetV2** trained on MRL Eye. It outputs `p_eye_closed`, and the specialist test accuracy is around **98.6%**.
For the mouth module, the selected model is **ResNet18** trained on YawDD mouth crops. It outputs `p_yawn`, and the specialist test accuracy is around **99.4%**.
The third component is not a neural network. It is a rule-based temporal fusion layer.
This separation is intentional: the CNNs detect visual evidence, while the rules decide whether the evidence is persistent enough to become a warning-candidate state.

中文脚本

VisionGuard 中每个模块都负责一种可见行为。
眼部模块选择的是在 MRL Eye 数据集上训练的 **MobileNetV2**，它输出 `p_eye_closed`，也就是闭眼概率，专门模型的测试准确率大约是 **98.6%**。
嘴部模块选择的是在 YawDD 嘴部裁剪图上训练的 **ResNet18**，它输出 `p_yawn`，也就是打哈欠概率，专门模型的测试准确率大约是 **99.4%**。
第三个部分不是神经网络，而是基于规则的时间融合层。
这种分离是有意设计的：CNN 负责识别视觉证据，而规则层负责判断这些证据是否持续到足以形成预警候选状态。

Page 8, 30 seconds

Training Protocol

English script

For training, we used transfer learning with ImageNet-pretrained CNN backbones, including ResNet18, MobileNetV2, and EfficientNet-B0.
The training process followed a two-stage strategy. First, the backbone was frozen so that only the classifier head was trained. Then, the full network was fine-tuned.
We used standard preprocessing such as resizing to 224 by 224 and ImageNet normalization.
Weighted cross entropy was used to handle class imbalance, especially for the yawn class.

中文脚本

在训练阶段，我们使用了基于 ImageNet 预训练模型的迁移学习，测试的 backbone 包括 ResNet18、MobileNetV2 和 EfficientNet-B0。
训练过程分为两个阶段。第一阶段先冻结 backbone，只训练分类头。第二阶段再对整个网络进行 fine-tuning。
输入图像会统一 resize 到 224 × 224，并使用 ImageNet normalization。
由于打哈欠类别数量相对较少，我们使用 weighted cross entropy 来缓解类别不平衡问题。

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Model Evidence Results

English script

This slide compares specialist test accuracy across the three CNN baselines.
For the mouth task, ResNet18 achieved the strongest test result, slightly ahead of EfficientNet-B0 and MobileNetV2, so it was selected for the mouth/yawn module.
For the eye task, MobileNetV2 gave the best overall balance between accuracy, macro F1, closed-eye recall, and real-time efficiency, so it was selected as the primary eye model.
The key point is that these numbers are specialist model results. They should not be described as final system-level drowsiness accuracy.

中文脚本

这一页比较了三个 CNN baseline 在专门任务上的测试准确率。
对于嘴部任务，ResNet18 的测试结果最好，略高于 EfficientNet-B0 和 MobileNetV2，所以它被选为嘴部和打哈欠模块的最终模型。
对于眼部任务，MobileNetV2 在准确率、macro F1、闭眼召回率和实时效率之间取得了最好的平衡，所以它被选为主要眼部模型。
这里最关键的是，这些数值是专门模型的结果，不能说成整个系统最终的疲劳检测准确率。

Page 10, 45 seconds

Temporal Decision Fusion

English script

This slide explains how warning states emerge over time.
The eye signal uses rolling evidence, because a very short eye closure may simply be a blink. The mouth signal also uses temporal context, because one isolated mouth movement should not immediately trigger a strong warning.
The fusion layer checks persistence and co-occurrence. For example, a high-confidence warning candidate is produced only when eye-warning evidence and recent yawning evidence overlap or appear close together in time.
Signal quality is also handled separately. If the face is not visible, the system treats it as an unreliable signal, not as drowsiness evidence.

中文脚本

这一页解释预警状态是如何随着时间形成的。
眼部信号会使用 rolling evidence，因为非常短暂的闭眼可能只是眨眼。嘴部信号也会考虑时间上下文，因为单独一次嘴部动作不应该立刻触发强预警。
融合层会检查证据是否持续，以及不同证据是否在时间上共同出现。例如，只有当眼部预警证据和近期打哈欠证据在时间上重叠或接近时，系统才会产生 high-confidence warning candidate。
同时，信号质量也会被单独处理。如果人脸不可见，系统会把它判断为 signal unreliable，而不是直接当成疲劳证据。

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Working System Interface

English script

This is the working interface of VisionGuard.
The Live Monitor page shows the camera area, eye and yawn warning counts, the Drowsiness Risk gauge, the current-session waveform, and recent events.
The interface is designed to make the system understandable, not just technically functional. Users can see not only that a warning appeared, but also what type of evidence contributed to it.
The same system also supports video upload analysis, history review, and insights, which makes the prototype closer to a complete monitoring workflow.

中文脚本

这一页展示的是 VisionGuard 的实际系统界面。
Live Monitor 页面包含摄像头区域、闭眼和打哈欠预警次数、Drowsiness Risk 仪表盘、当前 session 的波形图，以及最近事件列表。
这个界面的目标不只是让系统“能运行”，而是让系统结果更容易理解。用户不仅能看到是否出现预警，也能看到这个预警主要来自哪一种证据。
同一个系统还支持视频上传分析、历史记录查看和 Insights 分析，所以它更接近一个完整的监测工作流。

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From Evidence to Awareness

English script

To conclude, VisionGuard turns visual evidence into driver awareness.
It does this through specialist models, temporal fusion, and an explainable interface.
Now we will move to the live demo: first Live Monitor, then Video Upload Analysis, and finally History and Insights.

中文脚本

最后总结一下，VisionGuard 的核心价值是把视觉证据转化为驾驶员可理解的风险提醒。
它通过专门模型、时间融合逻辑和可解释界面来完成这个过程。
接下来我们进入 live demo：先展示实时监测，然后展示视频上传分析，最后展示历史记录和 Insights 页面。

Strict speaking boundary

During the presentation, do **not** say:

Our system achieves 99% drowsiness detection accuracy.

Say this instead:

Our specialist mouth/yawn classifier achieved about 99.4% test accuracy, and our specialist eye model achieved about 98.6% test accuracy. The full system uses these signals to generate rule-based warning-candidate states.